

## Mobile Input Documentation

This documentation provides an overview of the mobile input system. The system utilizes the **Enhanced Touch API** of the Unity Input System to handle complex RTS gestures like formation dragging, multi-selection, and camera control. To use it simply add the **Mobile Input** Prefab to your scene. It can be found under Large Scale Combat System / Prefabs.

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### 1. Input Gesture Reference

The MobileInputManager handles all touch interactions and delegates actions to the FormationController and CameraController.

| Gesture           | Mode      | Action                                                                                    |
|-------------------|-----------|-------------------------------------------------------------------------------------------|
| Single Tap        | Any       | Select unit (friendly), Order Attack (enemy), or Order Movement (ground).                 |
| 1-Finger Drag     | Standard  | <b>Camera Pan:</b> Moves the camera view across the battlefield.                          |
| 1-Finger Drag     | Selection | <b>Box Selection:</b> Zoids a selection rectangle to select multiple units.               |
| Long Press + Drag | Standard  | <b>Formation Drag:</b> Adjusts the width and rotation of the selected units' destination. |
| 2-Finger Pinch    | Any       | <b>Zoom:</b> Pinch in or out to change camera height.                                     |
| 2-Finger Twist    | Any       | <b>Camera Rotation:</b> Rotate the camera yaw by twisting two fingers.                    |

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### 2. Interaction Modes

#### Standard Mode (Default)

In this mode, the focus is on camera navigation and unit movement.

- **Panning:** A quick drag moves the camera based on panSensitivity.
- **Formation Adjustment:** By holding a finger down for a set duration (longPressTime), the player enters the "Formation Drag" state. This allows drawing the exact line/width the formation should occupy at the destination.

#### Selection Mode (Toggle)

Activated via the **Selection Mode Toggle Button**.

- While active, the button highlights with the `selectedColor`.
  - Dragging a finger will draw a **Selection Box** on the screen.
  - The system converts screen pixel coordinates into canvas local coordinates to ensure the visual box matches the touch position exactly.
  - Units within the box are added to the selection list in the `FormationController`.
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### 3. UI Elements

- **Selection Mode Button:** Toggles between camera panning and box selection logic.
  - **Deselect Button (X):** This button automatically appears only when at least one unit is selected. Pressing it clears all current selections and hides the button.
  - **Selection Box Visual:** A UI Image that scales and moves to provide visual feedback during box selection. It is automatically disabled when the touch is released.
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### 4. Technical Implementation Details

#### Coordinate System Fixes

**Note:** To ensure the selection box follows the finger accurately on different screen resolutions, the system uses `RectTransformUtility.ScreenPointToLocalPointInRectangle`. It also requires the UI Image pivot to be set to (0,0) (Bottom-Left).

#### Component Requirements

1. **EnhancedTouchSupport:** The `MobileInputManager` automatically enables `EnhancedTouchSupport` in `OnEnable()` and disables it in `OnDisable()`.
2. **Input Action References:** Requires two Action References for **Touch Position** and **Touch Contact** (Press).
3. **Canvas Settings:** The selection box logic accounts for the `Canvas.scaleFactor` to maintain accuracy regardless of UI scaling.